

Independent Study

AIEA Lab task 2

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Objectives

- Choosing research direction for the quarter
- Future plan and plan for next week

Direction/Proposal

I would like to research the cost/environmental efficiency of using modern neurosymbolic pipelines on logic benchmarks vs. increasing reasoning effort on reasoning models. I hope to see neurosymbolic approaches on lower reasoning effort models outperform pure high-reasoning effort models on a standard pipeline.

Motivation

I was looking at a research paper called [U-Math](#). U-Math is a dataset with university math problems. The paper tests different models on this dataset. The thing that stood out to me was inference vs. reasoning models. Here is the line that stood out to me: “Reasoning models attain the top U-MATH, U-MATHT, and U-MATHV scores of 86.8%, 93.1% and 58.5% respectively, compared to 67.2%, 71.7% and 47.0% for the standard-inference models.” Reasoning models outperform inference models every time.

LRM's

Large reasoning models, as described in this [video by IBM](#), are models that are fine-tuned to solve logic and math-type problems. They also undergo a reinforcement learning feedback loop. Small model judges judge cot pipelines created by the LRM. This process is repeated until the judge models are happy with the response. These models are much better at logical tasks. However, it comes at heavy computational costs, latency, and potential environmental impacts.

LRM's and Neurosymbolic AI

LRMs can be valuable to neurosymbolic pipelines, potentially contributing to translations from NL to FOL. However, my goal is to prove that increasing reasoning effort is negligible and consumes large computational power compared to deploying neurosymbolic pipelines alongside lower reasoning effort LRMs.

Conclusion/Next Week

Someday there should be true neurosymbolic LLMs the way LRMs exist. LLMs fine-tuned on logical translation and robustness can be interesting, and I might do something similar in this project. For next week I will explore LRMs and NeSy papers.

